Exercise Sheet 9

All Haar measures are normalised to total mass one. You may use the Weyl integration formula of Lecture 9. For a compact connected group K with maximal torus T and Weyl group W, and a continuous class function f,

$$\int_K f\,dk = \frac{1}{|W|}\int_T f(t)\,J(t)\,dt, \qquad J(t) = \prod_{\alpha\in R}(1-t^\alpha) = \prod_{\alpha\in R^+}|1-t^\alpha|^2.$$

On $\mathrm{SU}(2)$ we write $t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. Then the formula reads

$$\int f\,dk = \frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta.$$

The remark on indicator functions in Lecture 9 extends the formula to indicators of sets $\{c \geq 0\}$, where c is a continuous real class function.

Exercise 9.1 (Root planes in $\mathrm{SU}(3)$). Let $K = \mathrm{SU}(3)$ and $T = \{t = \mathrm{diag}(z_1, z_2, z_3) : |z_j| = 1,\ z_1z_2z_3 = 1\}$. Let E_{jk} be the matrix unit.

1. Show that $\mathrm{Ad}_t\, E_{jk} = z_j z_k^{-1} E_{jk}$ for $j \neq k$.
2. Let α be the root with $t^\alpha = z_1\bar{z}_2$. Put $X_\alpha = E_{12} - E_{21}$ and $Y_\alpha = i(E_{12} + E_{21})$. Show that X_α and Y_α lie in $\mathfrak{su}(3)$. Show that Ad_t rotates $\mathrm{span}_{\mathbb{R}}\{X_\alpha, Y_\alpha\}$ through θ, where $t^\alpha = e^{i\theta}$.
3. Deduce that $J(t) = \prod_{j<k}|z_j - z_k|^2$.
4. Let $\omega = e^{2\pi i/3}$. Evaluate J at $t_1 = \mathrm{diag}(1, \omega, \omega^2)$ and at $t_2 = \mathrm{diag}(-1, -1, 1)$. Find $\dim Z_K(t_2)$ and compare with the formula $\dim Z_K(t) = \dim T + 2\#\{\alpha > 0 : t^\alpha = 1\}$.

Exercise 9.2 ($\mathrm{SU}(2)$ as the 3-sphere). Write $g \in \mathrm{SU}(2)$ as

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1,$$

and identify g with $x = (x_0, x_1, x_2, x_3) = (\mathrm{Re}\,a, \mathrm{Im}\,a, \mathrm{Re}\,b, \mathrm{Im}\,b) \in S^3 \subset \mathbb{R}^4$.

1. Show that left and right multiplication by $h \in \mathrm{SU}(2)$ extend to orthogonal linear maps of $\mathbb{R}^4$. Conclude that normalised surface measure σ on S^3 is the Haar measure.
2. Show that g is conjugate to t_θ, with $\theta \in [0, \pi]$, exactly when $\mathrm{Re}\,a = \cos\theta$. Describe this conjugacy class.
3. Use the parametrisation $x = (\cos\theta, \sin\theta\, u)$ with $u \in S^2$ and $\theta \in [0, \pi]$. Show that $\sigma(\{\theta_1 \leq \theta \leq \theta_2\}) = \frac{2}{\pi}\int_{\theta_1}^{\theta_2}\sin^2\theta\,d\theta$. Compare with the Weyl formula.

Exercise 9.3 (Characters and multiplicities for $\mathrm{SU}(2)$). Let $\chi_m(t_\theta) = \sin(m+1)\theta/\sin\theta$ be the character of V_m.

1. Show that $\int_{\mathrm{SU}(2)}\chi_m\chi_n\,dk = \delta_{mn}$.
2. Compute the multiplicity of V_2 in $V_1^{\otimes 4}$ by integration.
3. Using weights only, find the full decomposition of $V_1^{\otimes 4}$. Check it by dimensions. Compare with part (b) and with the multiplicity 2 of V_0 found by integration in Lecture 9.
4. Compute $\int_{\mathrm{SU}(2)}\chi_2^3\,dk$. Interpret the result.

Exercise 9.4 (The group $\mathrm{SO}(3)$). Let

$$L_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and $R_\varphi = \exp(\varphi L_z)$. The torus is $T = \{R_\varphi\}$. For $v \in \mathbb{R}^3$ let $\hat{v}$ be the matrix with $\hat{v}w = v \times w$. Then $\hat{e}_1 = L_x$, $\hat{e}_2 = L_y$ and $\hat{e}_3 = L_z$.

1. Show that $R\hat{v}R^{-1} = \widehat{Rv}$ for $R \in \mathrm{SO}(3)$. Deduce that Ad_{R_φ} rotates $\mathfrak{m} = \mathrm{span}\{L_x, L_y\}$ through φ, and that $J(R_\varphi) = 4\sin^2(\varphi/2)$.
2. Let $w_0 = \mathrm{diag}(1, -1, -1)$. Show that $w_0 R_\varphi w_0^{-1} = R_{-\varphi}$. Use $|W| = 2$ to derive
$$\int_{\mathrm{SO}(3)} f \, dk = \frac{1}{\pi} \int_0^{\pi} f(R_\varphi)(1 - \cos\varphi)\, d\varphi.$$
3. Show that the standard representation of $\mathrm{SO}(3)$ on $\mathbb{C}^3$ is irreducible.
4. Find the Haar probability that a rotation has angle at most $\pi/2$.
5. Explain why $\sin(\varphi/2)$ is not a function on T, while J is.

Exercise 9.5 (The group $\mathrm{U}(2)$). Let $K = \mathrm{U}(2)$ and $T = \{\mathrm{diag}(e^{ia}, e^{ib})\}$. Recall from Lecture 9 that $\mathcal{X}(T) = \{i\ell : \ell \in X^*(T)\}$. Identify $\mathcal{X}(T)$ with $\mathbb{Z}^2$, where (p, q) labels the weight μ with $t^\mu = e^{i(pa+qb)}$. The roots are $\pm(1, -1)$ and $|W| = 2$.

1. Write down J. Show that $\rho \notin \mathcal{X}(T)$, while $D(t) = 1 - e^{-i(a-b)}$ is a function on T with $|D|^2 = J$.
2. Write the Weyl formula for class functions on $\mathrm{U}(2)$. Compute $\int_{\mathrm{U}(2)} |\operatorname{tr} g|^2 dk$ and $\int_{\mathrm{U}(2)} |\operatorname{tr} g|^4 dk$.
3. Interpret both answers in terms of $\mathbb{C}^2$ and $\mathbb{C}^2 \otimes \mathbb{C}^2$.

Exercise 9.6 (The constant $|W|$ for $\mathrm{U}(n)$ and $\mathrm{SU}(n)$). *Optional extension.* Use the Vandermonde identity

$$\prod_{j<k} (z_j - z_k) = \det\bigl(z_j^{n-k}\bigr)_{j,k=1}^{n} = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma)\, z^{e_\sigma}, \qquad e_\sigma = (n - \sigma(1), \dots, n - \sigma(n)).$$

1. Let T be the diagonal torus of $\mathrm{U}(n)$. Show that $\int_T \prod_{j<k} |z_j - z_k|^2 \, dt = n!$.
2. Show the same for the diagonal torus of $\mathrm{SU}(n)$, where $z_1 \cdots z_n = 1$.
3. Relate both results to the lemma $\int_T J \, dt = |W|$ of Lecture 9.